

Lecture 03: Linking Relocatable ELF File

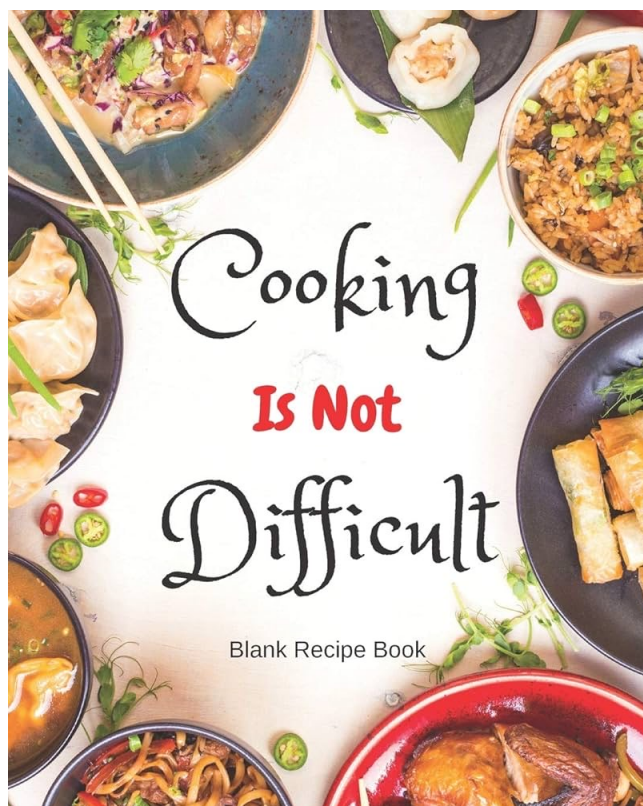
Vivek Kumar

Computer Science and Engineering

IIIT Delhi

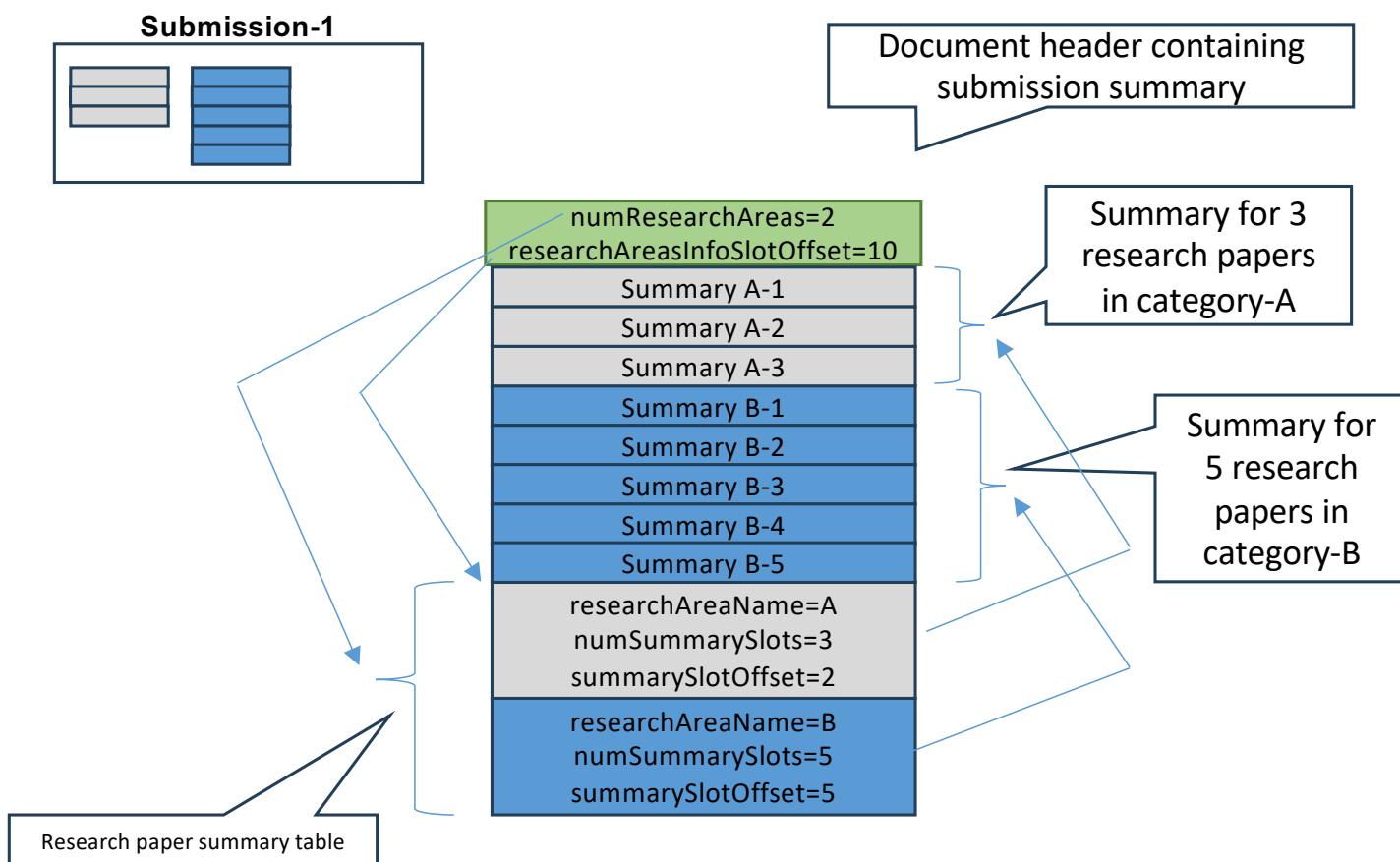
vivekk@iiitd.ac.in

Last Lecture: The Recipe book



- Who is the cook?
 - The one who executes the service
 - E.g., the Operating System!
- What is the recipe?
 - A well-structured format for recording recipe that the cook can understand
 - E.g., the executable file format
- The index page in the book
 - Header in the executable file format that helps in browsing the book
- The ingredients in the recipe?
 - Program data stored in the executable file format
- The red sauce
 - Statically linked reference if you download its recipe and permanently attach to your recipe book
 - Dynamically linked reference if you look it up online only while actually cooking
- Who prepares the recipe book??
 - Its you who is playing the role of a compiler!

Last Lecture: Sample Assignment



- I proposed a specific layout to overcome this challenge, and to make it easy to understand student's document
- I can easily “**load**” the student “**file**” from any number of students in my “**memory**” if they follow a standard “**format**” to store information, e.g., as the one shown below for the **Submission-1** shown here

The Story So Far...

- A **cook** (*OS*) manages a **kitchen** (*computer*)
 - Lecture 01
- The job of a cook is to do **cooking** (*execute binaries*) by reading **recipes** (*executable file*) in a **standard format** (*ELF*)
 - Lecture 02
- The cook has received recipe of a dish in **different files** (*relocatable files*). It will have to use a **stapler** (*linker*) to stitch these files to make a final recipe
 - **Today's point of discussion!**

Today's Class

- Linking relocatable ELF files
 - Symbol resolution
- Revisiting Executable File Format (ELF)

ELF Header (The index page in Book)

```
#define EI_NIDENT 16
```

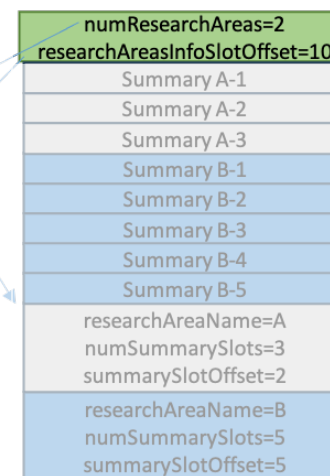
```
typedef struct {
    unsigned char    e_ident[EI_NIDENT];
    Elf32_Half       e_type;
    Elf32_Half       e_machine;
    Elf32_Word       e_version;
    Elf32_Addr       e_entry;
    Elf32_Off        e_phoff;
    Elf32_Off        e_shoff;
    Elf32_Word       e_flags;
    Elf32_Half       e_ehsize;
    Elf32_Half       e_phentsize;
    Elf32_Half       e_phnum;
    Elf32_Half       e_shentsize;
    Elf32_Half       e_shnum;
    Elf32_Half       e_shstrndx;
} Elf32_Ehdr;
```

```
typedef struct {
    short numResearchAreas;
    short researchAreasInfoSlotOffset;
} CSE231_Ehdr; // All three summarized?
               // Column index
```

```
vivek@possum:~/os23$ readelf -h ./a.out
ELF Header:
  Magic:   7f 45 4c 46 02 01 01 00 00 00 00 00 00 00 00 00
  Class:             ELF64
  Data:              2's complement, little endian
  Version:           1 (current)
  OS/ABI:             UNIX - System V
  ABI Version:       0
  Type:              DYN (Position-Independent Executable file)
  Machine:           Advanced Micro Devices X86-64
  Version:           0x1
  Entry point address: 0x1160
  Start of program headers: 64 (bytes into file)
  Start of section headers: 14320 (bytes into file)
  Flags:              0x0
  Size of this header: 64 (bytes)
  Size of program headers: 56 (bytes)
  Number of program headers: 13
  Size of section headers: 64 (bytes)
  Number of section headers: 31
  Section header string table index: 30
```



Document header containing submission summary



Summary for 3 research papers in category-A

Summary for 5 research papers in category-B

Research paper summary table

Image source: <https://refspecs.linuxfoundation.org/elf/gabi4+/ch4.eheader.html>

ELF Header

Identification
for object file

Object
file type

```
#define EI_NIDENT 16
```

```
typedef struct {
    unsigned char    e_ident[EI_NIDENT];
    Elf32_Half       e_type;
    Elf32_Half       e_machine;
    Elf32_Word       e_version;
    Elf32_Addr       e_entry;
    Elf32_Off        e_phoff;
    Elf32_Off        e_shoff;
    Elf32_Word       e_flags;
    Elf32_Half       e_ehsize;
    Elf32_Half       e_phentsize;
    Elf32_Half       e_phnum;
    Elf32_Half       e_shentsize;
    Elf32_Half       e_shnum;
    Elf32_Half       e_shstrndx;
} Elf32_Ehdr;
```

Entry point address for
starting the executable

Program Header Table (PHT) file offset

Section Header Table (SHT) file offset

ELF header size in bytes (fixed)

Size in bytes of each entry in PHT

Total number of entries in PHT

Size in bytes of each entry in SHT

Total number of entries in SHT

Index of the section called as
"String Table"

Image source: <https://refspecs.linuxfoundation.org/elf/gabi4+/ch4.eheader.html>

Entry Point?

```
vivek@possum:~/elf_os-12$ readelf -h ./a.out
ELF Header:
  Magic:   7f 45 4c 46 01 01 01 00 00 00 00 00 00 00 00 00
  Class:                                ELF32
  Data:                                      2's complement, little endian
  Version:                               1 (current)
  OS/ABI:                                UNIX - System V
  ABI Version:                           0
  Type:                                  DYN (Shared object file)
  Machine:                               Intel 80386
  Version:                               0x1
  Entry point address:                   0x3e0
  Start of program headers:              52 (bytes into file)
  Start of section headers:              6060 (bytes into file)
  Flags:                                  0x0
  Size of this header:                   52 (bytes)
  Size of program headers:               32 (bytes)
  Number of program headers:              9
  Size of section headers:               40 (bytes)
  Number of section headers:             29
  Section header string table index:     28
```

Entry point
address

Entry point basically provides the location of the first instruction that has to be executed (?)

Entry Point?

```
vivek@possum:~/elf_os-12$ readelf -h ./a.out
ELF Header:
  Magic:   7f 45 4c 46 01 01 01 00 00 00 00 00 00 00 00 00
  Class:                                ELF32
  Data:                                      2's complement, little endian
  Version:                               1 (current)
  OS/ABI:                                UNIX - System V
  ABI Version:                           0
  Type:                                  DYN (Shared object file)
  Machine:                               Intel 80386
  Version:                               0x1
  Entry point address:                   0x3e0
  Start of program headers:              52 (bytes into file)
  Start of section headers:              6060 (bytes into file)
  Flags:                                  0x0
  Size of this header:                   52 (bytes)
  Size of program headers:               32 (bytes)
  Number of program headers:              9
  Size of section headers:               40 (bytes)
  Number of section headers:             29
  Section header string table index:     28
vivek@possum:~/elf_os-12$ objdump -d ./a.out | grep -A16 3e0
000003e0 <_start>:
3e0: 31 ed                xor    %ebp,%ebp
3e2: 5e                  pop    %esi
3e3: 89 e1                mov    %esp,%ecx
3e5: 83 e4 f0             and    $0xfffffffff0,%esp
3e8: 50                  push   %eax
3e9: 54                  push   %esp
3ea: 52                  push   %edx
3eb: e8 22 00 00 00       call   412 <_start+0x32>
3f0: 81 c3 e8 1b 00 00    add    $0x1be8,%ebx
3f6: 8d 83 f8 e5 ff ff    lea    -0x1a08(%ebx),%eax
3fc: 50                  push   %eax
3fd: 8d 83 98 e5 ff ff    lea    -0x1a68(%ebx),%eax
403: 50                  push   %eax
404: 51                  push   %ecx
405: 56                  push   %esi
406: ff b3 20 00 00 00    pushl  0x20(%ebx)
40c: e8 af ff ff ff      call   3c0 <__libc_start_main@plt>
```

Entry point
actual method

Entry point
address

Entry point
address

_start calls
__libc_start_main
with the address of
user main as a
parameter (invoked
from here)

ELF Section Header (Chapter Details)

```
typedef struct {
    Elf32_Word  sh_name;
    Elf32_Word  sh_type;
    Elf32_Word  sh_flags;
    Elf32_Addr  sh_addr;
    Elf32_Off   sh_offset;
    Elf32_Word  sh_size;
    Elf32_Word  sh_link;
    Elf32_Word  sh_info;
    Elf32_Word  sh_addralign;
    Elf32_Word  sh_entsize;
} Elf32_Shdr;
```

ELF_Shdr does not store name directly

```
typedef struct {
    char* researchAreaName;
    short numSummarySlots;
    short summarySlotOffset;
} CSE231_Shdr;
```

// "A" or "B" or "C" ?
 // Total papers summarized
 // Index to its first summary line

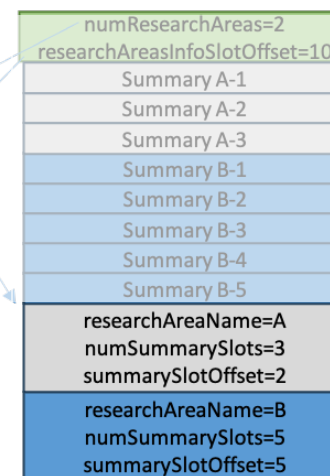
```
vivek@possum:~/elf_os-l2$ readelf -S hello.o
There are 17 section headers, starting at offset 0x3d4:
```

[Nr]	Name	Type	Addr	Off	Size	ES	Flg	Lk	Inf	Al
[0]		NULL	00000000	000000	000000	00		0	0	0
[1]	.group	GROUP	00000000	000034	000008	04		14	16	4
[2]	.text	PROGBITS	00000000	00003c	000059	00	AX	0	0	1
[3]	.rel.text	REL	00000000	0002e4	000040	08		I 14	2	4
[4]	.data	PROGBITS	00000000	000098	000004	00	WA	0	0	4
[5]	.bss	NOBITS	00000000	00009c	000000	00	WA	0	0	1
[6]	.rodata	PROGBITS	00000000	00009c	000025	00	A	0	0	1
[7]	.data.rel.local	PROGBITS	00000000	0000c4	000004	00	WA	0	0	4
[8]	.rel.data.rel.loc	REL	00000000	000324	000008	08		I 14	7	4
[9]	.text.__x86.get_p	PROGBITS	00000000	0000c8	000004	00	AXG	0	0	1
[10]	.comment	PROGBITS	00000000	0000cc	00002a	01	MS	0	0	1
[11]	.note.GNU-stack	PROGBITS	00000000	0000f6	000000	00		0	0	1
[12]	.eh_frame	PROGBITS	00000000	0000f8	000060	00	A	0	0	4
[13]	.rel.eh_frame	REL	00000000	00032c	000010	08		I 14	12	4
[14]	.symtab	SYMTAB	00000000	000158	000130	10		15	12	4
[15]	.strtab	STRTAB	00000000	000288	00005b	00		0	0	1
[16]	.shstrtab	STRTAB	00000000	00033c	000096	00		0	0	1

Key to Flags:
 W (write), A (alloc), X (execute), M (merge), S (strings), I (info),
 L (link order), O (extra OS processing required), G (group), T (TLS),
 C (compressed), x (unknown), o (OS specific), E (exclude),
 p (processor specific)



Document header containing submission summary



Summary for 3 research papers in category-A

Summary for 5 research papers in category-B

Research paper summary table

ELF Section Header

```
typedef struct {
    Elf32_Word
    Elf32_Word
    Elf32_Word
    Elf32_Addr
    Elf32_Off
    Elf32_Word
    Elf32_Word
    Elf32_Word
    Elf32_Word
    Elf32_Word
} Elf32_Shdr;
```

Offset (bytes) into the section "String Table" where the name of this section is stored

```
sh_name;
sh_type;
sh_flags;
sh_addr;
sh_offset;
sh_size;
sh_link;
sh_info;
sh_addralign;
sh_entsize;
```

Section type (symbol table, string table, programmer defined data, etc.)

Runtime information for section, e.g., writable data, executable data, loadable

Address at which first byte of this section should reside at runtime (actual address in PHT for a.out)

Offset (bytes) of the starting byte in this section from the beginning of the object file

Size of the section (vary across sections)

How Linker **Reads** Section Names?

- The section header does not contains the name of its corresponding section
 - It's structure has a member that contains an offset into the ELF file where the name of this section is stored
- **The names of all the sections are stored inside .shstrtab section**
- Each section header stores the offset into the .shstrtab section (not the SHDR but the actual section) where the name of that section is stored as a null terminated string

Motivating Example

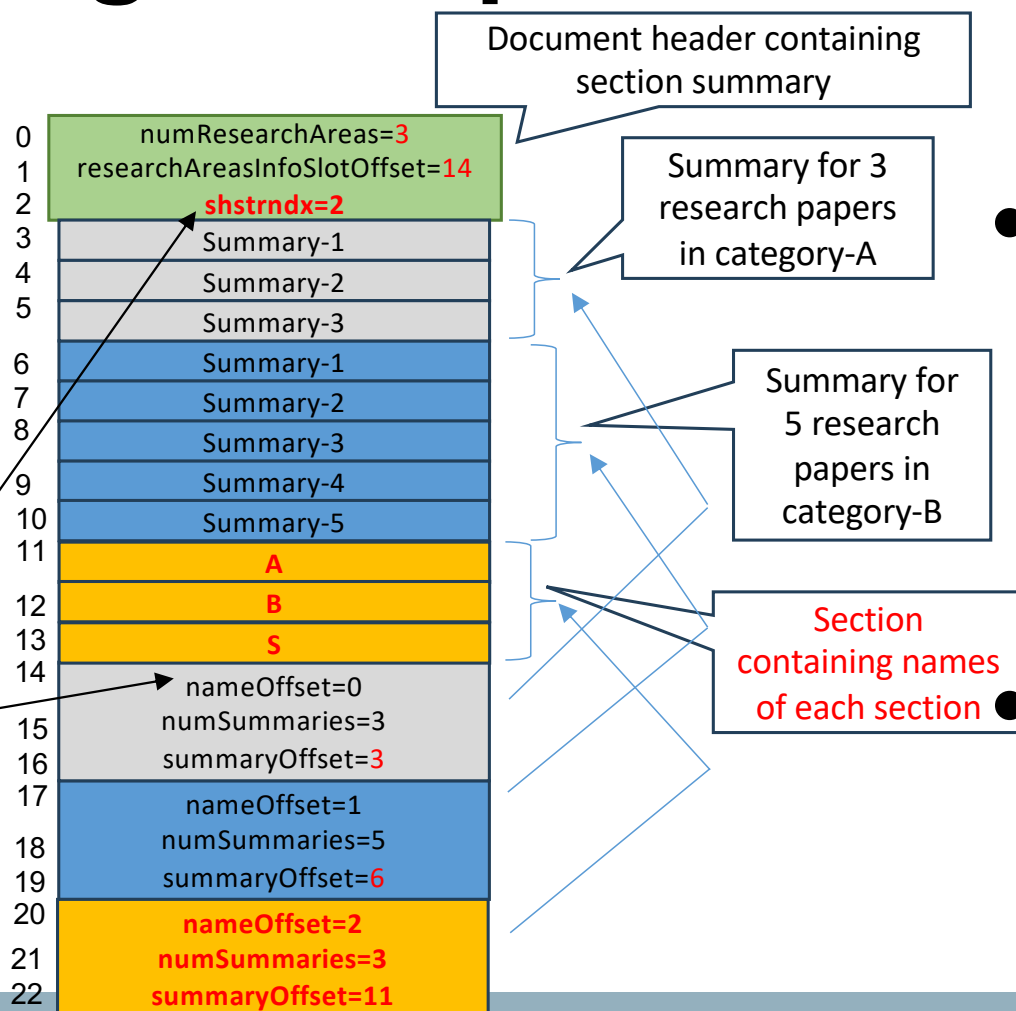
Submission-1



```
#define EI_NIDENT 16

typedef struct {
    unsigned char  e_ident[EI_NIDENT];
    Elf32_Half     e_type;
    Elf32_Half     e_machine;
    Elf32_Word     e_version;
    Elf32_Addr     e_entry;
    Elf32_Off      e_phoff;
    Elf32_Off      e_shoff;
    Elf32_Word     e_flags;
    Elf32_Half     e_ehsize;
    Elf32_Half     e_phentsize;
    Elf32_Half     e_phnum;
    Elf32_Half     e_shentsize;
    Elf32_Half     e_shnum;
    Elf32_Half     e_shstrndx;
} Elf32_Ehdr;

typedef struct {
    Elf32_Word     sh_name;
    Elf32_Word     sh_type;
    Elf32_Word     sh_flags;
    Elf32_Addr     sh_addr;
    Elf32_Off      sh_offset;
    Elf32_Word     sh_size;
    Elf32_Word     sh_link;
    Elf32_Word     sh_info;
    Elf32_Word     sh_addralign;
    Elf32_Word     sh_entsize;
} Elf32_Shdr;
```



- It is the same motivational example we discussed in Lecture 02
 - Changes shown in **orange** and **red color**
- We added a new section “S” that contains the name of all the sections in this document
 - Document header contains the index of this particular section summary
 - Name of this section (“S”) will also be now stored in the actual content of this section

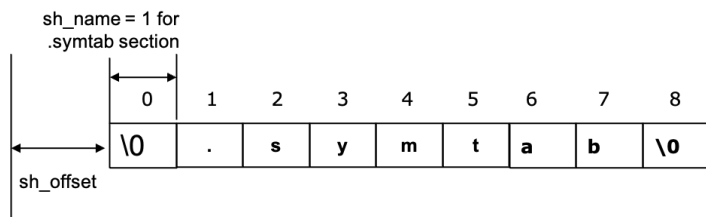
Each section summary now also contains **offset from the start of the content of section “S”**, where the name is stored

The “.shstrtab” Section in ELF

```
vivek@possum:~/elf_os-l2$ readelf -p .shstrtab hello

String dump of section '.shstrtab':
[ 1] .symtab
[ 9] .strtab
[11] .shstrtab
[1b] .interp
[23] .note.ABI-tag
[31] .note.gnu.build-id
[44] .gnu.hash
[4e] .dynsym
[56] .dynstr
[5e] .gnu.version
[6b] .gnu.version_r
[7a] .rel.dyn
[83] .rel.plt
[8c] .init
[92] .plt.got
[9b] .text
[a1] .fini
[a7] .rodata
[af] .eh_frame_hdr
[bd] .eh_frame
[c7] .init_array
[d3] .fini_array
[df] .dynamic
[e8] .data
[ee] .bss
[f3] .comment
```

- It is a String Table Section
- Stores the name of each sections in the binary
 - String (NULL terminated)
- As with other sections, this section is also a contiguous chunk of memory



First column is the offset (hex value) from the start for the first character of the corresponding section name. First entry starts at index 1

Few Other Interesting Sections

```
vivek@possum:~/elf_os-l2$ readelf -S hello
There are 29 section headers, starting at offset 0x17fc:

Section Headers:
 [Nr] Name                Type              Addr      Off      Size    ES Flg Lk Inf Al
 [ 0] .interp                PROGBITS          00000000 000000 000000 00  A  0  0  0
 [ 1] .note.ABI-tag          NOTE              00000154 000154 000013 00  A  0  0  1
 [ 2] .note.gnu.build-id     NOTE              00000168 000168 000020 00  A  0  0  4
 [ 3] .gnu.hash              GNU_HASH          00000188 000188 000024 00  A  0  0  4
 [ 4] .gnu.hash              GNU_HASH          000001ac 0001ac 000020 04  A  5  0  4
 [ 5] .dynsym                DYNSYM            000001cc 0001cc 000080 10  A  6  1  4
 [ 6] .dynstr                STRTAB            0000024c 00024c 00009d 00  A  0  0  1
 [ 7] .gnu.version           VERSYM            000002ea 0002ea 000010 02  A  5  0  2
 [ 8] .gnu.version_r         VERNEED           000002fc 0002fc 000030 00  A  6  1  4
 [ 9] .rel.dyn               REL               0000032c 00032c 000048 08  A  5  0  4
[10] .rel.plt               REL               00000374 000374 000010 08  AI 5 22 4
[11] .init                  PROGBITS          00000384 000384 000023 00  AX 0  0  4
[12] .plt                   PROGBITS          000003b0 0003b0 000030 04  AX 0  0 16
[13] .plt.got               PROGBITS          000003e0 0003e0 000010 08  AX 0  0  8
[14] .text                  PROGBITS          000003f0 0003f0 000202 00  AX 0  0 16
[15] .fini                  PROGBITS          000005f4 0005f4 000014 00  AX 0  0  4
[16] .rodata                PROGBITS          00000608 000608 00002d 00  A  0  0  4
[17] .eh_frame_hdr          PROGBITS          00000638 000638 00003c 00  A  0  0  4
[18] .eh_frame              PROGBITS          00000674 000674 0000fc 00  A  0  0  4
[19] .init_array            INIT_ARRAY        00001ed8 000ed8 000004 04  WA 0  0  4
[20] .fini_array            FINI_ARRAY        00001edc 000edc 000004 04  WA 0  0  4
[21] .dynamic                DYNAMIC           00001ee0 000ee0 0000f8 08  WA 6  0  4
[22] .got                   PROGBITS          00001fd8 000fd8 000028 04  WA 0  0  4
[23] .data                  PROGBITS          00002000 001000 000010 00  WA 0  0  4
[24] .bss                   NOBITS            00002020 001010 400020 00  WA 0  0 32
[25] .comment               PROGBITS          00000000 001010 000029 01  MS 0  0  1
[26] .symtab                SYMTAB            00000000 00103c 000460 10  27 43 4
[27] .strtab                STRTAB            00000000 00149c 000264 00  0  0  1
[28] .shstrtab              STRTAB            00000000 001700 0000fc 00  0  0  1

Key to Flags:
W (write), A (alloc), X (execute), M (merge), S (strings), I (info),
L (link order), O (extra OS processing required), G (group), T (TLS),
C (compressed), x (unknown), o (OS specific), E (exclude),
p (processor specific)
```

```
vivek@possum:~/elf_os-l2$ cat hello.c
#include<stdio.h>

int my_global1 = 1, my_global2[1024*1024];
char* str = "Updated value of my_global1";

int main() {
    my_global1 += 1;
    printf("%s = %d\n", str, my_global1);
    return 0;
}
```


Executable Instructions (“`.text`”)

```
vivek@possum:~/elf_os-12$ objdump -d hello.o | grep -A20 text
Disassembly of section .text:

00000000 <main>:
 0: 8d 4c 24 04      lea    0x4(%esp),%ecx
 4: 83 e4 f0         and    $0xffffffff0,%esp
 7: ff 71 fc         pushl  -0x4(%ecx)
 a: 55              push   %ebp
 b: 89 e5           mov    %esp,%ebp
 d: 53              push   %ebx
 e: 51              push   %ecx
 f: e8 fc ff ff ff  call   10 <main+0x10>
14: 05 01 00 00 00   add    $0x1,%eax
19: 8b 90 00 00 00   mov    0x0(%eax),%edx
1f: 83 c2 01         add    $0x1,%edx
22: 89 90 00 00 00   mov    %edx,0x0(%eax)
28: 8b 88 00 00 00   mov    0x0(%eax),%ecx
2e: 8b 90 00 00 00   mov    0x0(%eax),%edx
34: 83 ec 04         sub    $0x4,%esp
37: 51              push   %ecx
38: 52              push   %edx
39: 8d 90 1c 00 00   lea    0x1c(%eax),%edx
```

```
vivek@possum:~/elf_os-12$ objdump -d hello | grep -A20 text
Disassembly of section .text:

000003f0 <_start>:
3f0: 31 ed           xor    %ebp,%ebp
3f2: 5e             pop    %esi
3f3: 89 e1           mov    %esp,%ecx
3f5: 83 e4 f0         and    $0xffffffff0,%esp
3f8: 50             push   %eax
3f9: 54             push   %esp
3fa: 52             push   %edx
3fb: e8 22 00 00 00   call   422 <_start+0x32>
400: 81 c3 d8 1b 00   add    $0x1bd8,%ebx
406: 8d 83 18 e6 ff   lea    -0x19e8(%ebx),%eax
40c: 50             push   %eax
40d: 8d 83 b8 e5 ff   lea    -0x1a48(%ebx),%eax
413: 50             push   %eax
414: 51             push   %ecx
415: 56             push   %esi
416: ff b3 20 00 00   pushl  0x20(%ebx)
41c: e8 af ff ff ff   call   3d0 <__libc_start_main@
421: f4             hlt
```

- Stores bytes corresponding to program execution (user code and the GNU C library supporting code)
- The above output shows the content of “`.text`” section for relocatable v/s executable
 - ‘`A20`’ prints 20 lines after a match is found by `grep` command

Section “.rodata”

```
vivek@possum:~/elf_os-12$ cat hello.c
#include<stdio.h>

int my_global1 = 1, my_global2[1024*1024];
char* str = "Updated value of my_global1";

int main() {
    my_global1 += 1;
    printf("%s = %d\n", str, my_global1);
    return 0;
}
```

```
vivek@possum:~/elf_os-12$ readelf -p .rodata hello

String dump of section '.rodata':
[      8] Updated value of my_global1
[     24] %s = %d^J
```

- Stores all the string literals defined in the program
 - First column in readelf output is the offset from the start for that particular string
- Question
 - Would the content change across relocatable and executable?

Section “.data”

```
vivek@possum:~/elf_os-12$ cat hello.c
#include<stdio.h>

int my_global1 = 1, my_global2[1024*1024];
char* str = "Update value of my_global1";

int main() {
    my_global1 += 1;
    printf("%s = %d\n", str, my_global1);
    return 0;
}
```

```
vivek@possum:~/elf_os-12$ readelf -S hello.o | grep '\.data \| Name'
[Nr] Name                Type           Addr      Off      Size    ES Flg Lk Inf Al
[ 4] .data                 PROGBITS       00000000  000098  000004  00  WA  0  0  4
```

- Contains the **initialized value** of all the global/static variables from the user code (not the variable name)
 - In our running example it is the **value** of the variable “my_global1”
 - Why not the value of “str”?
- “WA” flag indicates that the section content is to be loaded at runtime in memory and is writable
- PROGBITS indicating it contains information defined by the user code
- Where is the name of the global variable stored?

Sections “.symtab” and “.strtab”

```
vivek@possum:~/elf_os-12$ readelf -p .symtab hello.o

String dump of section '.symtab':
[   d4]
[   da] @
[   f0] #
[   f8] Y
[  100] (
[  110] >
[  120] T

vivek@possum:~/elf_os-12$ readelf -p .strtab hello.o

String dump of section '.strtab':
[    1] hello.c
[    9] my_global1
[   14] my_global2
[   1f] str
[   23] main
[   28] __x86.get_pc_thunk.ax
[   3e] _GLOBAL_OFFSET_TABLE_
[   54] printf
```

- “.symtab” is like a directory to map the name of the symbols (present inside .strtab) with the actual content (value) of that symbol
 - E.g., for the symbol “my_global1”, symbol table “.symtab” will contain an index into the string table “.strtab” where the name “my_global1” is stored

Section “.bss”

```
vivek@possum:~/elf_os-12$ cat hello.c
#include<stdio.h>

int my_global1 = 1, my_global2[1024*1024];
char* str = "Updated value of my_global1";

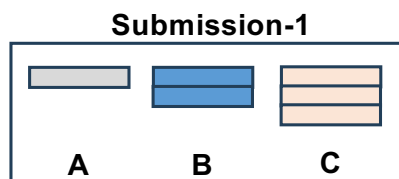
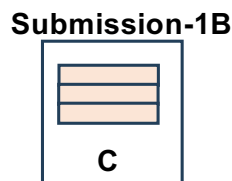
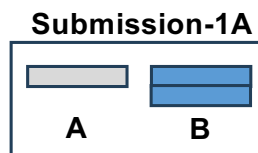
int main() {
    my_global1 += 1;
    printf("%s = %d\n", str, my_global1);
    return 0;
}
```

Hex

```
vivek@possum:~/elf_os-12$ readelf -S hello | grep '\.bss' | Name'
[Nr] Name Type Addr Off Size ES Flg Lk Inf Al
[24] .bss NOBITS 00002020 001010 400020 00 WA 0 0 32
```

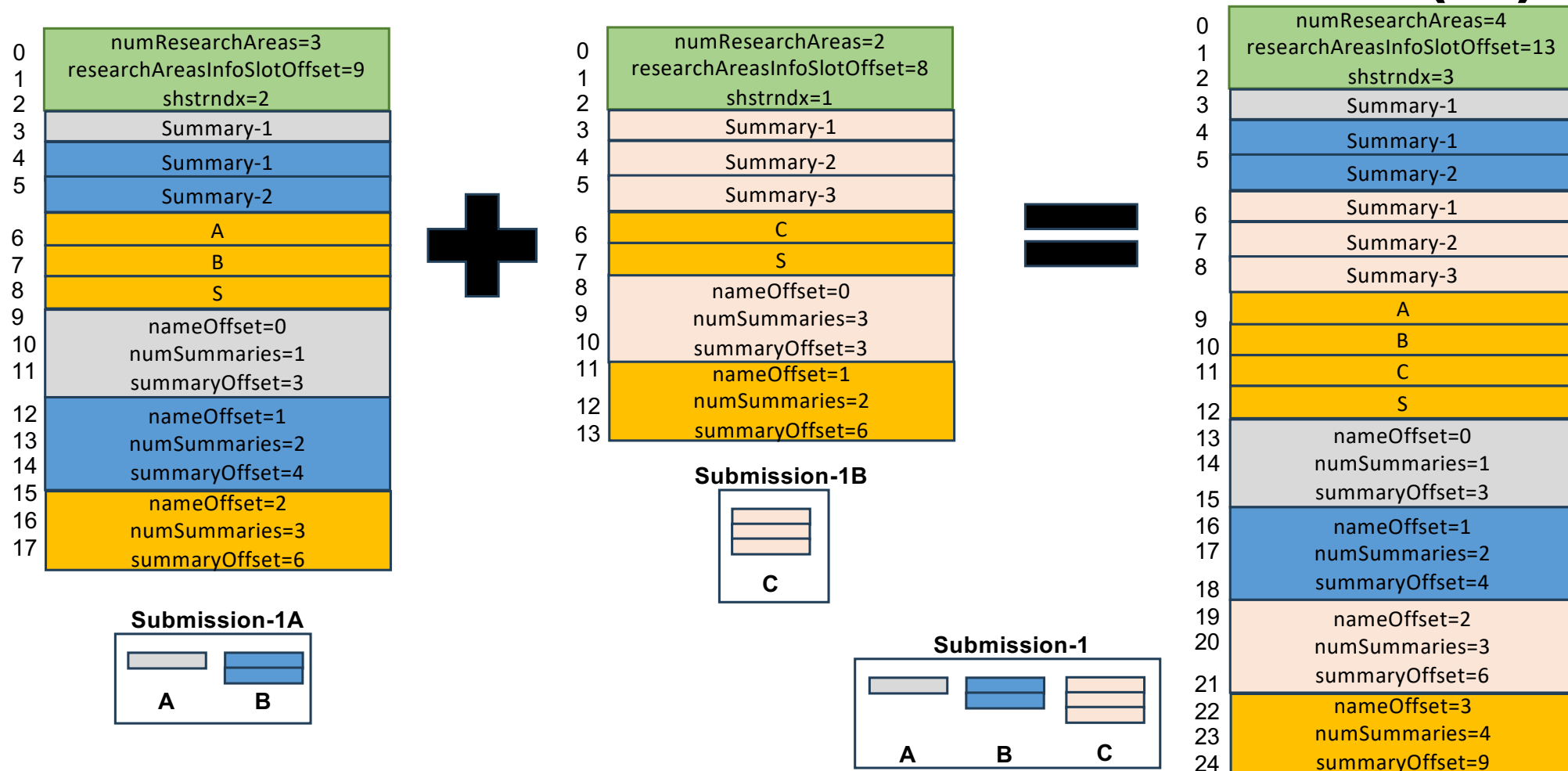
- .bss (Block Started by Symbol) section indicates the total memory to be allocated at runtime for **uninitialized** global/static variables
 - In our running example it is the **value** of the variable “my_global2”
- “WA” flag indicates that the section content is to be loaded at runtime in memory and is writable
- NOBITS indicating it relates to the user code, but it doesn't occupy any space in the object file

How Linker **Stitches** Two Relocatable (.o) ?

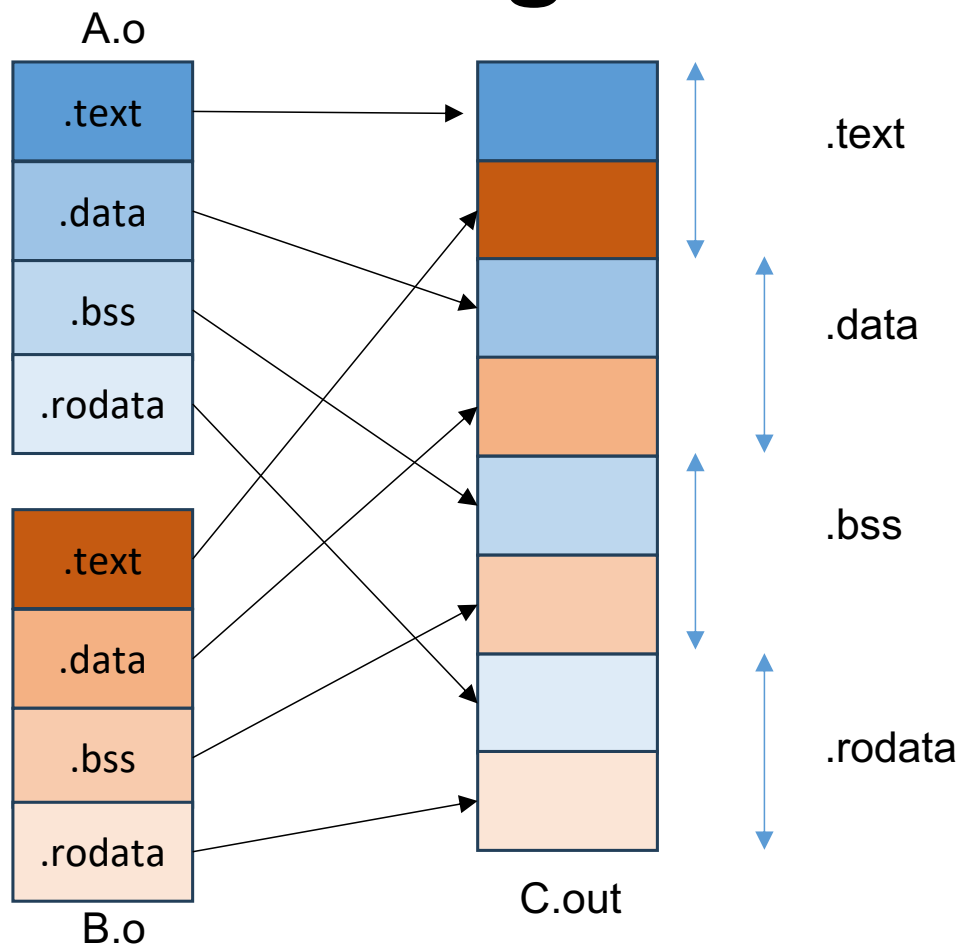


- Let us revisit our motivational example of student assignment to summarize the research areas in a single line
- I announced upfront that there must be a single submission from each student, but one student submitted two separate files BUT in correct format (our modified ELF)
 - Submission-1A having summaries for areas A and B
 - Submission-1B having summary for area C
- I asked the student to **STITCH** both the submissions into a single submission (the job of a linker!)
 - Submission-1 having summaries for A, B, and C

How Linker **Stitches** Two Relocatable (.o) ?



Linker: Merges the Sections



- **Relocation** is the process of merging the sections and resolving the symbol addresses

Linker: Relocation

```
[vivek@possum]$ cat fib.c
int get_number();

int fib(int n) {
    if(n<2) return n;
    else return fib(n-1)+fib(n-2);
}

int compute() {
    int n = get_number();
    return fib(n);
}
```

```
[vivek@possum]$ cat call-fib.c
int compute();
int get_number() {
    return 40;
}

int main() {
    int result = compute();
    return 0;
}
```

Compilation

fib.o

SYM Name	Type	Location
get_number	External	NULL
fib	Internal	Addr_A
compute	Internal	Addr_B

call-fib.o

SYM Name	Type	Location
compute	External	NULL
get_number	Internal	Addr_C

Resolve symbols by linking
fib.o and call-fib.o together

a.out

SYM Name	Location
get_number	Addr_D
fib	Addr_E
compute	Addr_F

Sections at Linking and Executable

```
vivek@possum:~/elf_os-12$ readelf -S hello.o
There are 17 section headers, starting at offset 0x3d4:
```

[Nr]	Name	Type	Addr	Off	Size	ES	Flg	Lk	Inf	Al
[0]		NULL	00000000	000000	000000	00		0	0	0
[1]	.group	GROUP	00000000	000034	000008	04		14	16	4
[2]	.text	PROGBITS	00000000	00003c	000059	00		AX	0	1
[3]	.rel.text	REL	00000000	0002e4	000040	08		I 14	2	4
[4]	.data	PROGBITS	00000000	000098	000004	00		WA	0	4
[5]	.bss	NOBITS	00000000	00009c	000000	00		WA	0	1
[6]	.rodata	PROGBITS	00000000	00009c	000025	00		A	0	1
[7]	.data.rel.local	PROGBITS	00000000	0000c4	000004	00		WA	0	4
[8]	.rel.data.rel.loc	REL	00000000	000324	000008	08		I 14	7	4
[9]	.text.____x86.get_p	PROGBITS	00000000	0000c8	000004	00		AXG	0	1
[10]	.comment	PROGBITS	00000000	0000cc	00002a	01		MS	0	1
[11]	.note.GNU-stack	PROGBITS	00000000	0000f6	000000	00			0	1
[12]	.eh_frame	PROGBITS	00000000	0000f8	000060	00		A	0	4
[13]	.rel.eh_frame	REL	00000000	00032c	000010	08		I 14	12	4
[14]	.symtab	SYMTAB	00000000	000158	000130	10		15	12	4
[15]	.strtab	STRTAB	00000000	000288	00005b	00			0	1
[16]	.shstrtab	STRTAB	00000000	00033c	000096	00			0	1

Key to Flags:
W (write), A (alloc), X (execute), M (merge), S (strings), I (info),
L (link order), O (extra OS processing required), G (group), T (TLS),
C (compressed), x (unknown), o (OS specific), E (exclude),
p (processor specific)

Q: Why there are more number of sections in executable v/s relocatable file?

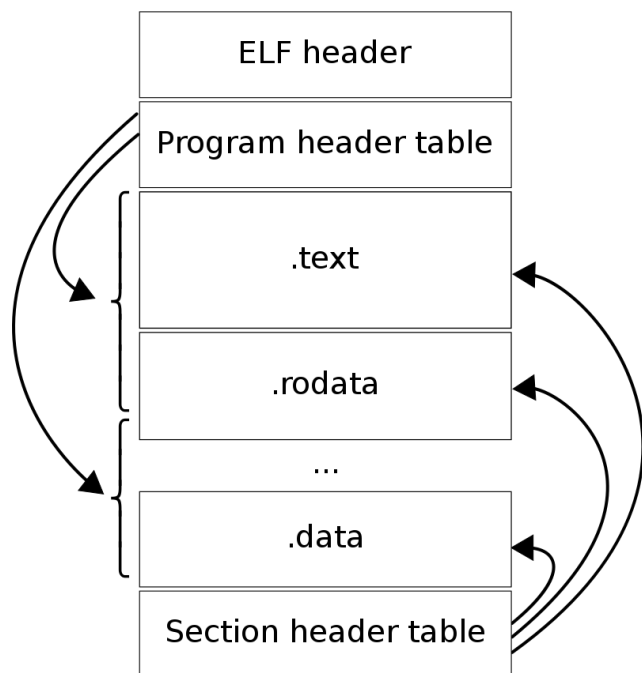
A: To assist the dynamic loader, the linker adds several more sections in the executable than in the relocatable

```
vivek@possum:~/elf_os-12$ readelf -S hello
There are 29 section headers, starting at offset 0x17fc:
```

[Nr]	Name	Type	Addr	Off	Size	ES	Flg	Lk	Inf	Al
[0]		NULL	00000000	000000	000000	00		0	0	0
[1]	.interp	PROGBITS	00000154	000154	000013	00		A	0	1
[2]	.note.ABI-tag	NOTE	00000168	000168	000020	00		A	0	4
[3]	.note.gnu.build-id	NOTE	00000188	000188	000024	00		A	0	4
[4]	.gnu.hash	GNU_HASH	000001ac	0001ac	000020	04		A	5	4
[5]	.dynsym	DYNSYM	000001cc	0001cc	000080	10		A	6	1
[6]	.dynstr	STRTAB	0000024c	00024c	00009d	00		A	0	1
[7]	.gnu.version	VERSYM	000002ea	0002ea	000010	02		A	5	2
[8]	.gnu.version_r	VERNEED	000002fc	0002fc	000030	00		A	6	1
[9]	.rel.dyn	REL	0000032c	00032c	000048	08		A	5	4
[10]	.rel.plt	REL	00000374	000374	000010	08		AI	5	22
[11]	.init	PROGBITS	00000384	000384	000023	00		AX	0	4
[12]	.plt	PROGBITS	000003b0	0003b0	000030	04		AX	0	16
[13]	.plt.got	PROGBITS	000003e0	0003e0	000010	08		AX	0	8
[14]	.text	PROGBITS	000003f0	0003f0	000202	00		AX	0	16
[15]	.fini	PROGBITS	000005f4	0005f4	000014	00		AX	0	4
[16]	.rodata	PROGBITS	00000608	000608	00002d	00		A	0	4
[17]	.eh_frame_hdr	PROGBITS	00000638	000638	00003c	00		A	0	4
[18]	.eh_frame	PROGBITS	00000674	000674	0000fc	00		A	0	4
[19]	.init_array	INIT_ARRAY	00001ed8	000ed8	000004	04		WA	0	4
[20]	.fini_array	FINI_ARRAY	00001edc	000edc	000004	04		WA	0	4
[21]	.dynamic	DYNAMIC	00001ee0	000ee0	0000f8	08		WA	6	4
[22]	.got	PROGBITS	00001fd8	000fd8	000028	04		WA	0	4
[23]	.data	PROGBITS	00002000	001000	000010	00		WA	0	4
[24]	.bss	NOBITS	00002020	001010	400020	00		WA	0	32
[25]	.comment	PROGBITS	00000000	001010	000029	01		MS	0	1
[26]	.symtab	SYMTAB	00000000	00103c	000460	10			27	43
[27]	.strtab	STRTAB	00000000	00149c	000264	00			0	1
[28]	.shstrtab	STRTAB	00000000	001700	0000fc	00			0	1

Key to Flags:
W (write), A (alloc), X (execute), M (merge), S (strings), I (info),
L (link order), O (extra OS processing required), G (group), T (TLS),
C (compressed), x (unknown), o (OS specific), E (exclude),
p (processor specific)

Executable and Linkable Format (ELF)



Note: For simplicity, we will only discuss ELF format for 32-bit object files in CSE231

Image source: https://en.wikipedia.org/wiki/Executable_and_Linkable_Format

- **ELF header**
 - Provides a roadmap for the entire file organization
 - Always at offset zero of the object file
 - Provides entry point address for execution
- **Two views of an ELF file**
 - Linkable view (relocatable file)
 - Execution view (executable file, shared object)
- **Linkable view**
 - Section header table
 - One section header for each section
 - Sections
 - Contains data required for linking
 - Machine code, global variables, (initialized and un-initialized), symbol tables, line mapping between machine code and original C code, etc.
- **Execution (or Loader) view**
 - Program header table
 - One program header for each segment
 - Segments
 - Created by merging several sections
 - Contains information required for by the loader for execution
- **Contiguous chunk of memory (ELF header, PHT, SHT, each section, each segment)**

Segments in Executable File

```
vivek@possum:~/elf_os-l2$ readelf -l hello

Elf file type is DYN (Shared object file)
Entry point 0x3f0
There are 9 program headers, starting at offset 52

Program Headers:
  Type           Offset             VirtAddr           PhysAddr          FileSiz MemSiz  Flg Align
  PHDR            0x00000034          0x00000034          0x00000034         0x00120 0x00120  R   0x4
  INTERP          0x00000154          0x00000154          0x00000154         0x00013 0x00013  R   0x1
    [Requesting program interpreter: /lib/ld-linux.so.2]
  LOAD            0x00000000          0x00000000          0x00000000         0x00770 0x00770  R E 0x1000
  LOAD            0x00000ed8          0x000001ed8          0x000001ed8         0x00138 0x400168 RW 0x1000
  DYNAMIC          0x00000ee0          0x000001ee0          0x000001ee0         0x000f8 0x000f8  RW  0x4
  NOTE            0x00000168          0x000000168          0x000000168         0x00044 0x00044  R   0x4
  GNU_EH_FRAME    0x00000638          0x000000638          0x000000638         0x0003c 0x0003c  R   0x4
  GNU_STACK       0x00000000          0x00000000          0x00000000         0x00000 0x00000  RW  0x10
  GNU_RELRO       0x00000ed8          0x000001ed8          0x000001ed8         0x00128 0x00128  R   0x1

Section to Segment mapping:
Segment Sections...
 00
 01      .interp
 02      .interp .note.ABI-tag .note.gnu.build-id .gnu.hash .dynsym .dynstr .
gnu.version .gnu.version_r .rel.dyn .rel.plt .init .plt .plt.got .text .fini .
rodata .eh_frame_hdr .eh_frame
 03      .init_array .fini_array .dynamic .got .data .bss
 04      .dynamic
 05      .note.ABI-tag .note.gnu.build-id
 06      .eh_frame_hdr
 07
```

- Segments are created by merging several sections and each segment contains a specific type of information required for by the loader for execution (i.e., all the sections inside a segment are related)
 - Note the RE Flag for PHDR index 2 v/s RW Flag for index 3
- Not available in relocatable file

Summary: Portability with ELF

- Recall, one of the challenges of OS is to support portability
- Machine code is architecture dependent
- Having an architecture independent layout of object files helps in achieving portability
 - The OS will have same set of steps to load executables across different processor architectures (portability)
 - Easier to change and test the tools used for manipulating object files

Reference Materials

- Executable and Linkable Format

- <https://man7.org/linux/man-pages/man5/elf.5.html>
- <https://www.sco.com/developers/devspecs/gabi41.pdf>
- https://wiki.osdev.org/ELF_Tutorial

Next Lecture

- Loading ELF Executable & procedure calling convention